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SEMICONDUCTOR ELECTRONIC DEVICE WITH EDGE TERMINATION REGION AND MANUFACTURING PROCESS

Abstract

A semiconductor electronic device has a semiconductor body that has a first conductivity type, a front surface and a rear surface at a distance from the front surface along a first direction. The semiconductor body also has a lateral edge. The device has an active area which accommodates, in use, a conductive channel of the device; and an edge termination region around the active area, between the active area and the lateral edge of the semiconductor body along a second direction transversal to the first direction. The edge termination region has a plurality of doped portions of a second conductivity type different from the first conductivity type and arranged in the semiconductor body at a distance from each other along the first direction, or along the first direction and along the second direction.

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Background/Summary

PRIORITY CLAIM

[0001] This application claims the priority benefit of Italian Application for U.S. Pat. No. 10,202,4000004948, filed on Mar. 6, 2024, the content of which is hereby incorporated by reference in its entirety to the maximum extent allowable by law.

TECHNICAL FIELD

[0002] The present invention relates to a semiconductor electronic device and, in particular, to a vertical-conduction device, with an edge termination region, and to the manufacturing process for a vertical-conduction device.

BACKGROUND

[0003] As is known, semiconductor electronic devices (for example MOSFET transistors) used in power applications need to withstand high voltages, for example even higher than 400 V.

[0004] In power applications, a high breakdown voltage and a low on-state resistance are parameters desired in order to improve the electrical performance of electronic devices.

[0005] FIG. 1 shows a MOSFET device **1** based on silicon, for power applications, in a Cartesian reference system XYZ.

[0006] The MOSFET device **1** is a vertical-conduction device and has a superjunction-type structure (based on a charge compensation principle) which allows the MOSFET device **1** to overcome the classic limits of silicon, that is to obtain a good trade-off between high breakdown voltage and low on-state resistance.

[0007] The MOSFET device **1** is formed in a die comprising an active area **2** and an edge region **3**.

[0008] The MOSFET device **1** comprises a semiconductor body **4** made of silicon, having a front surface **4A** and a rear surface **4B**, and having an N-type doping. The semiconductor body **4** comprises a substrate **6** and an epitaxial region **8** grown on the substrate **6**.

[0009] The active area **2**, of which only a part is shown in FIG. 1, is configured to accommodate, in use, a conductive channel of the MOSFET device **1**.

[0010] The edge region **3** extends around the active area **2**, in a peripheral portion of the die, between the active area **2** and an outer edge **10** of the semiconductor body **4**.

[0011] The MOSFET device **1** comprises, in the active area **2**, a plurality of surface body regions **11**, having a P-type doping, which extend, from the front surface **4A**, inside the epitaxial region **8** along the Z axis.

[0012] The surface body regions **11** extend at a distance from each other along the X axis.

[0013] Source regions **12**, having an N-type doping, each extend inside a respective surface body region **11**.

[0014] Deep body regions **13**, having a P-type doping, extend in depth into the epitaxial region **8**, along the Z axis, each starting from a respective surface body region **11**. The presence of the deep body regions **13** defines the superjunction structure of the MOSFET device **1**.

[0015] An insulating region **15** extends onto the front surface **4A** of the semiconductor body **4** and gate insulated regions **16** are arranged inside the insulating region **15**, over the front surface **4A**.

[0016] Each gate insulated region **16** is arranged between two adjacent surface body regions **11**, partially overlying the surface body regions **11**.

[0017] A conductive region **18** extends into openings of the insulating region **15**, in contact with the

source regions **12**.

[0018] The edge region **3** comprises an edge termination region **20** having a P-type doping which extends at the surface in the epitaxial region **8**, starting from the front surface **4A**, and has the shape of a ring surrounding the active area **2**. The edge termination region **20** extends continuously along the X axis from the active area **2** towards the outer edge **10**. The edge termination region **20** has a width along the X axis comprised between 50 μm and 200 μm .

[0019] The MOSFET device **1** also has an equipotential ring **21** having an N-type doping, which extends at the surface in the epitaxial region **8**, at the outer edge **10** of the semiconductor body **4**, and a conductive region **22** which extends onto the front surface **4A** in contact with the equipotential ring **21**.

[0020] The superjunction structure allows high breakdown voltages of the MOSFET device **1** to be obtained, for example up to 1000 V and above. The presence of the ring region **20** contributes to increasing the breakdown voltage of the MOSFET device **1**.

[0021] However, the presence of the ring region **20** requires additional manufacturing steps (lithography, implantation and diffusion) with respect to those leading to the formation of the surface **11** and deep **13** body regions. Accordingly, the MOSFET device **1** has high manufacturing costs.

[0022] FIG. **2** shows another example of a MOSFET device **50**, also a vertical-conduction and superjunction-type device. The MOSFET device **50** has the same structure as the MOSFET device **1** of FIG. **1**; accordingly, common elements are indicated by the same reference numerals.

[0023] In the MOSFET device **50**, the edge region **3** also comprises further edge termination regions forming a plurality of columns **51** having a P-type doping which extend in depth into the semiconductor body **4** starting from the front surface **4A**.

[0024] The columns **51** extend continuously along the Z axis, from the front surface **4A** through the ring region **20** and the epitaxial region **8**.

[0025] However, the MOSFET device **50** also has high manufacturing costs.

[0026] There is a need in the art to provide a power MOSFET which overcome the drawbacks noted above.

SUMMARY

[0027] In an embodiment, a semiconductor electronic device comprises: a semiconductor body having a first conductivity type, a front surface and a rear surface extending at a distance from the front surface along a first direction, the semiconductor body also having a lateral edge; an active area configured to accommodate, in use, a conductive channel of the semiconductor electronic device; and an edge termination region extending around the active area, between the active area and the lateral edge of the semiconductor body along a second direction transversal to the first direction. The edge termination region comprises a plurality of doped portions having a second conductivity type different from the first conductivity type and arranged in the semiconductor body at a distance from each other along the first direction, or along the first direction and along the second direction

[0028] In an embodiment, a process for manufacturing a semiconductor electronic device comprises: providing a wafer of semiconductor material having a first conductivity type, a front surface and a rear surface at a distance from the front surface along a first direction, the wafer further having a lateral edge; forming an active area configured to accommodate, in use, a conductive channel of the semiconductor electronic device; and forming an edge termination region extending around the active area, between the active area and the lateral edge of the wafer along a second direction transversal to the first direction. The step of forming an edge termination region comprises forming a plurality of doped portions, having a second conductivity type different from the first conductivity type, in the semiconductor body and at a distance from each other along the first direction, or along the first direction and along the second direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] For a better understanding of the present invention, embodiments thereof are now described, purely by way of non-limiting example, with reference to the attached drawings, wherein:

[0030] FIG. 1 shows a cross-section of a MOSFET device;

[0031] FIG. 2 shows a cross-section of a further MOSFET device;

[0032] FIG. 3 shows a top-plan view of a semiconductor electronic device, according to an embodiment;

[0033] FIG. 4 shows a cross-section of the device of FIG. 3, along a section line IV-IV of FIG. 3;

[0034] FIGS. 5A-5G show the device of FIG. 4, in subsequent manufacturing steps;

[0035] FIGS. 6 and 7 show cross-sections of semiconductor electronic devices according to other embodiments; and

[0036] FIGS. 8 and 9 show top-plan views of semiconductor electronic devices according to further embodiments.

DETAILED DESCRIPTION

[0037] FIGS. 3 and 4 show a semiconductor electronic device, hereinafter simply referred to as device **100**, in a Cartesian reference system XYZ having axes X, Y, Z orthogonal to each other.

[0038] The device **100** is a semiconductor device based on silicon; however, the device **100** may be a semiconductor device based on different material, for example silicon carbide or other simple or compound semiconductor material.

[0039] The device **100** is formed into a die **101** which may be obtained after a dicing step of a semiconductor wafer.

[0040] The die **101** comprises an edge **102** which physically delimits the die **101**. In practice, the edge **102** is a lateral edge which laterally delimits the die **101**.

[0041] The device **100** comprises an active area **103** and an edge region **104** which extends around the active area **103**.

[0042] In particular, the edge region **104** extends in structural continuity with the active area **103**.

[0043] Typically, the active area **103** may extend into a central portion of the die **101**, and the edge region in a peripheral portion of the die **101**.

[0044] In practice, the edge region **104** extends between the active area **103** and the edge **102** of the die **101** and is externally delimited by the edge **102**.

[0045] The active area **103** is configured to accommodate, in use, a conductive channel of the device **100**.

[0046] The edge region **104** may comprise functional elements for reducing or preventing crowding of the electric field lines outside the active area **103**, as better described and illustrated hereinafter with reference to FIG. 4. In practice, the edge region **104** is configured not to accommodate, in use, a conductive channel of the device **100**.

[0047] As shown in the cross-section of FIG. 4, the device **100** may be a MOSFET device, in particular a vertical-conduction MOSFET device, even more in particular a superjunction MOSFET device.

[0048] The device **100** may be an electronic device for power applications, in particular for high voltages, for example higher than 400 V.

[0049] FIG. 4 shows only a portion of the active area **103**, in particular a portion of the active area **103** in proximity to the edge region **104**. The dash-dot line that separates the active area **103** from the edge region **104** is to be understood as qualitative.

[0050] The device **100** comprises a semiconductor body **110**, in particular of silicon, or of silicon carbide or other semiconductor material, having an N-type doping, and having a front surface **110A** and a rear surface **110B** at a distance from each other along the Z axis.

[0051] The front surface **110A** forms the front side of the semiconductor body **110** and delimits it upwardly. The rear surface **110B** forms the back side of the semiconductor body **110** and delimits it downwardly.

[0052] In practice, the front surface **110A** and the rear surface **110B** are opposite to each other along the Z axis.

[0053] The semiconductor body **110** also has a lateral edge which extends at a distance from the active area **103** along the X and Y axes. In practice, the lateral edge may physically delimit the semiconductor body **110**, for example on planes transversal to those of the front and rear surfaces **110A**, **110B**. The lateral edge of the semiconductor body **110** may coincide with the edge **102** of the die **101**, therefore it is indicated below with the same reference numeral.

[0054] In the embodiment shown, the semiconductor body **110** comprises a substrate **111** and a structural region **112** extending onto the substrate **111**. For example, the structural region **112** may be grown epitaxially on the substrate **111**; accordingly, hereinafter the structural region **112** is also referred to as an epitaxial region **112**.

[0055] The epitaxial region **112** forms, in the active area **103**, a drift region of the device **100**.

[0056] Substrate **111** and epitaxial region **112** may be of the same semiconductor material and have the same conductivity type (here N-type).

[0057] The substrate **111** may have a higher doping level than the epitaxial region **112**.

[0058] The device **100** comprises, in the active area **103**, surface body regions **115** having a P-type doping; deep body regions **116** having a P-type doping; source regions **117** having an N-type doping; and gate regions **118**.

[0059] For simplicity, in FIG. 4, only two surface body regions **115**, two deep body regions **116**, two source regions **117**, and two gate regions **118** are shown. However, it will be clear that the device **100** may comprise, in the active area **103**, further surface body regions **115**, deep body regions **116**, source regions **117**, and gate regions **118**.

[0060] The surface body regions **115** extend from the front surface **110A** into the semiconductor body **110**, in particular into the epitaxial region **112**, at a distance from each other along the X axis.

[0061] The source regions **117** extend, at least in part, each inside a respective surface body region **115**.

[0062] The deep body regions **116** extend, along the Z axis, in depth into the semiconductor body **110**, in particular into the epitaxial region **112**, each starting from a respective surface body region **115**.

[0063] The deep body regions **116** extend at a distance from the front surface **110A**, in direct contact with the surface body regions **115**.

[0064] In practice, the deep body regions **116** extend into the semiconductor body **110** to a depth, measured from the front surface **110A** along the Z axis, greater than the depth, measured from the front surface **110A** along the Z axis, of the surface body regions **115**.

[0065] The deep body regions **116** may have a width (along the X axis in the section of FIG. 4) smaller than the width (along the X axis in the section of FIG. 4) of the surface body regions **115**.

[0066] The gate regions **118** comprise a gate insulating portion **119** and a gate conductive portion **120** and extend onto the front surface **110A** partially overlying the surface body regions **115** along the Z axis.

[0067] A drain region of conductive material, not shown here, extends at the rear surface **110B** forming a drain terminal D of the device **100**.

[0068] In practice, in use, the surface body regions **115** are configured to accommodate the conductive channel of the device **100**, thus forming a conductive path extending along the Z axis through the semiconductor body **110**, between the front surface **110A** and the rear surface **110B**, as indicated in an exemplary and schematic manner by a dashed arrow in FIG. 4.

[0069] An insulating region **123** extends onto the front surface **110A** of the semiconductor body **110**, for example both in the active area **103** and in the edge region **104**.

[0070] A conductive region **124** extends onto the semiconductor body **110**, in electrical contact with the source regions **117**. In practice, the conductive region **124** is a source contact region and forms a source terminal S of the device **100**. For example, the conductive region **124** may extend through openings in the insulating region **123**, throughout the thickness of the insulating region **123**, to the front surface **110A**.

[0071] The conductive region **124** may be used, for example at different sections of the device **100** not shown here, to also contact the surface body regions **115**, so as to obtain a body-source short circuit.

[0072] The device **100** comprises, in the edge region **104**, an edge termination region **130** which extends inside the semiconductor body **110** around the active area **103**.

[0073] The edge termination region **130** comprises a plurality of pockets **132** having a P-type doping which extend into the semiconductor body **110**, in particular inside the epitaxial region **112**.

[0074] The edge termination region **130** may have a width $W_{sub,t}$, measured along a direction (X axis in the section of FIG. 4) extending between the active area and the lateral edge **102**, comprised for example between 50 μm and 200 μm . Greater values of the width $W_{sub,t}$ may be useful for the device **100** to be capable of operating at high voltages.

[0075] In practice, the width $W_{sub,t}$ may be defined as the width, for example the maximum width, occupied as a whole by the pockets **132** inside the semiconductor body **110**, between the active area **103** and the lateral edge **102**.

[0076] The pockets **132** may have a doping level comprised, for example, between $5 \times 10^{15} \text{ at/cm}^3$ and $5 \times 10^{17} \text{ at/cm}^3$, in order to maximize the breakdown voltage.

[0077] The pockets **132** extend at a distance from each other along the Z axis. For example, the distance $W_{sub,v}$ between two adjacent pockets **132** along the Z axis may be comprised between 3 μm and 8 μm .

[0078] The pockets **132** also extend at a distance from each other along the X axis. For example, the distance $W_{sub,h}$ between two adjacent pockets **132** along the X axis may be comprised between 2 μm and 6 μm .

[0079] The pockets **132** may each have a width $W_{sub,p}$, along the X axis, comprised for example between 1 μm and 10 μm .

[0080] In this embodiment, the pockets **132** are distributed in the semiconductor body **110** such that the edge termination region **130** has a P-type pocket density (number of pockets per unit area or unit volume), at least as a first approximation (that is, except for process variability), uniform inside the width $W_{sub,t}$. In practice, the pockets **132** are all arranged, at least as a first approximation (that is, except for process variability), at the same mutual distance $W_{sub,h}$ along the X axis and $W_{sub,v}$ along the Z axis, and have at least as a first approximation the same width $W_{sub,p}$.

[0081] In detail, the plurality of pockets **132** comprise surface pockets **132A**, which extend into the semiconductor body **110** at the front surface **110A**, and deep (or buried) pockets **132B** which extend into the semiconductor body **110** at a distance from the front surface **110A**.

[0082] The surface pockets **132A** extend starting from the front surface **110A** towards the inside of the semiconductor body **110**, in particular towards the inside of the epitaxial region **112**.

[0083] The surface pockets **132A** extend at a distance from each other along the X axis. In particular, the surface pockets **132A** extend, at least as a first approximation (that is, except for process variability), all at the same distance from each other along the X axis.

[0084] The deep pockets **132B** extend into the semiconductor body **110**, in particular into the epitaxial region **112**, at a distance from the front surface **110A** of the semiconductor body **110**.

[0085] The deep pockets **132B** extend at a distance from each other along the X axis and along the Z axis.

[0086] In this embodiment, the deep pockets **132B** are arranged in groups extending along the Z

axis each underneath a respective surface pocket **132A**.

[0087] In particular, in the example of FIG. **4**, four deep pockets **132B** are arranged underneath each surface pocket **132A**.

[0088] In detail, in the embodiment of FIG. **4**, the deep pockets **132B** are aligned with each other along the Z axis and are also aligned along the Z axis with the respective surface pocket **132A**. However, the pockets **132** may face each other along the Z axis in whole or in part.

[0089] The uniform distribution of the deep pockets **132B**, in particular the fact of having the same mutual distance $W_{sub.h}$ along the X axis and the same mutual distance $W_{sub.v}$ along the Z axis, may ensure greater simplicity in the design and manufacturing of the device **100**.

[0090] As shown in FIG. **3**, the surface pockets **132A** extend continuously around the active area **103**. In other words, the surface pockets **132A** have a substantially circular shape and form a plurality of concentric rings which completely surround the active area **103**.

[0091] The deep pockets **132B** may have, in plan view (e.g., on planes parallel to the XY plane), equal or different shape with respect to the surface pockets **132A**.

[0092] According to one embodiment, the deep pockets **132B** (all or only some of them) may also extend continuously around the active area **103**; this may allow the device **100** to obtain, in use, a high breakdown voltage.

[0093] The presence of the surface pockets **132A** may be considered optional, with the device including only the deep pockets **132B** spaced from the surface **110A** and positioned around the active area **103**.

[0094] The device **100** may comprise, optionally, an end region **140** arranged in the edge region **104** at the outer edge **102**.

[0095] The end region **140** comprises a doped region **141** having an N-type doping which extends into the semiconductor body **110**, at the front surface **110A**; and a conductive region **142** which extends onto the front surface **110A** in electrical contact with the doped region **141**. The end region **140** may have the function of forming an equipotential ring with the drain terminal of the device **100**, on the outer edge **102** of the die **101**.

[0096] The Inventor has verified that the plurality of pockets **132** that form the edge termination region **130**, at a distance from each other along the Z axis, allow, in use, a high breakdown voltage of the device **100** and therefore excellent electrical performance to be obtained.

[0097] In particular, the fact that the pockets **132** extend to a high depth into the semiconductor body **110**, for example between 20 μm and 80 μm , from the front surface **110A**, may ensure that the breakdown voltage of the device **100** is further increased.

[0098] In addition, the Inventor has verified that the fact that the pockets **132** extend, along the X axis, at a distance from each other along the X axis, contributes to increasing the electrical robustness, in use, of the device **100**. For example, the device **100** may have in use high breakdown voltages, higher than or equal to 1000 V.

[0099] Furthermore, the fact that the pockets **132** extend at a distance from each other along the X axis also allows the manufacturing costs of the device **100** to be decreased, in particular if the device **100** is a superjunction MOSFET device. In fact, as also discussed in detail below with reference to FIGS. **5A-5G**, the distance along the X axis, allows to avoid the use of an additional mask for the formation of a ring region which extends continuously along the X axis, as instead discussed for the known devices **1** and **50** of FIGS. **1** and **2**.

[0100] Hereinafter, with reference to FIGS. **5A** to **5G**, manufacturing steps of the device **100** are shown, with reference to the cross-section shown in FIG. **4**.

[0101] In FIG. **5A**, a wafer **200** of silicon is provided having a front surface **200A** and a rear surface **200B** opposite to the front surface **200A** parallel to the Z axis; alternatively, the wafer **200** may be of a different semiconductor material such as for example silicon carbide or other simple or compound semiconductor material.

[0102] The wafer **200** comprises the substrate, again indicated by **111**, on which an epitaxial layer

201 has been grown.

[0103] The epitaxial layer **201** and the substrate **111** are of the same material (silicon); however, they may be made of different materials.

[0104] The epitaxial layer **201** and the substrate **111** have the same conductivity type, here having an N-type doping.

[0105] The epitaxial layer **201** forms the front surface **200A** of the wafer **200**.

[0106] Furthermore, doped portions **203** having a P-type doping which will form the deep body regions **116**, and P-type doped portions which will form the deepest row of the deep pockets **132B** and therefore again indicated by **132B** have already been formed, in the wafer **200**.

[0107] For example, the doped portions **203**, **132B** may be formed by implanting P-type doping species into the epitaxial layer **201**, for example by using a lithographic mask to selectively implant the P-type doping species into the desired areas.

[0108] Subsequently, as shown in FIG. 5B, an epitaxial layer **205** is grown on the epitaxial layer **201**. The epitaxial layer **205** is of the same material (here silicon) as the epitaxial layer **201**; however, it may be of different material.

[0109] The epitaxial layer **205** forms a new front surface of the wafer **200**, again indicated by **200A** for simplicity. For clarity, in FIG. 5B, a dashed line divides the epitaxial layer **205** from the epitaxial layer **201**.

[0110] In FIG. 5C, doped portions **207** are formed in the epitaxial layer **205**, over and in contact with the doped portions **203**. For example, the doped portions **207** may be formed by implanting P-type doping species into the epitaxial layer **205**, for example by using a lithographic mask to selectively implant the P-type doping species into the desired areas.

[0111] In FIG. 5D, an epitaxial layer **209** is grown on the front surface **200A**, on the epitaxial layer **205**.

[0112] The epitaxial layer **209** is of the same material (here silicon) as the epitaxial layer **205**; however, it may be of different material.

[0113] The epitaxial layer **209** forms a new front surface of the wafer **200**, again indicated by **200A** for simplicity. For clarity, in FIG. 5D, a dashed line divides the epitaxial layer **205** from the epitaxial layer **209**.

[0114] In FIG. 5E, doped portions **211** having a P-type doping are formed in the epitaxial layer **209**, over and in contact with the doped portions **207**. Furthermore, P-type doped portions which will form a further row of deep pockets **132B**, therefore again indicated by **132B**, are also formed in the epitaxial layer **209**.

[0115] For example, the doped portions **211**, **132B** may be formed by implanting P-type doping species into the epitaxial layer **209**, for example by using a lithographic mask to selectively implant the P-type doping species into the desired areas.

[0116] In FIG. 5F, an epitaxial layer **213** is grown on the front surface **200A**, on the epitaxial layer **209**. The epitaxial layer **213** is of the same material (here silicon) as the epitaxial layer **209**; however, it may be of different material.

[0117] The epitaxial layer **213** forms a new front surface of the wafer **200**, again indicated by **200A** for simplicity. For clarity, in FIG. 5F, a dashed line divides the epitaxial layer **213** from the epitaxial layer **209**.

[0118] Following the growth of FIG. 5F, what has been described with reference to FIGS. 5C to 5E is repeated in sequence up to the formation of the surface pockets **132A** of the edge termination region **130** and the deep body regions **116** (FIG. 5G).

[0119] Furthermore, in FIG. 5G, the succession of epitaxial layers grown on top of each other forms the epitaxial region **112** of the device **100**.

[0120] In practice, the pockets **132** and the deep body regions **116** may be formed simultaneously by alternating epitaxial growth steps and doping species implantation steps.

[0121] The steps of FIGS. 5C to 5E are repeated a number of times that depends on the desired

thickness of the epitaxial region **112**, the desired thickness of the deep body regions **116**, and the desired number of pockets **132** in the edge termination region **130**.

[0122] One or more annealing steps may be performed to activate the pockets **132A**, **132B** and the deep body regions **116**.

[0123] Subsequently, in a manner not shown here, the remaining regions of the device **100** described with reference to FIG. **4** may be formed, including the surface body regions **115**, the source regions **117** and the gate regions **118**.

[0124] Finally, per se known manufacturing steps follow, such as dicing of the wafer **200** and formation of the electrical contacts, leading to the formation of the device **100**.

[0125] The described manufacturing process allows the manufacturing costs of the device **100** to be reduced, in particular when the device **100** is a MOSFET device with a superjunction structure. In fact, the pockets **132** of the edge termination region **130** may be formed with the same masks and the same implantation steps used for the formation of the deep body regions **116**, without additional dedicated lithography and implantation steps.

[0126] Furthermore, the described process may also have reduced complexity and execution times compared to the manufacturing processes of the known devices of FIGS. **1** and **2**.

[0127] FIG. **6** shows a different embodiment of the present electronic device, indicated here by **300**. The electronic device **300** has a general structure similar to that of the device **100**; accordingly, elements in common are indicated by the same reference numerals and are not further described in detail.

[0128] The device **300** is also a vertical-conduction MOSFET device of the superjunction type.

[0129] In detail, the device **300** comprises the semiconductor body **110** and has, in the active area **103**, the surface body regions **115**, the deep body regions **116**, the source regions **117** and the gate regions **118**.

[0130] The device **300** comprises, in the edge region **104**, an edge termination region **330** which extends inside the semiconductor body **110** around the active area **103**.

[0131] The edge termination region **330** also here comprises a plurality of pockets **332** of an opposite conductivity type (P) with respect to that of the semiconductor body **110** (N), including surface pockets **332A** and deep pockets **332B**, which extend into the semiconductor body **110**, in particular inside the epitaxial region **112**.

[0132] The edge termination region **330** may have the width $W_{sub.t}$ described with reference to the device **100**.

[0133] The pockets **332** may have, around the active area **103**, in plan view on planes parallel to the XY plane, the same trend as discussed with reference to the pockets **132** of FIG. **3**. For example, the pockets **332** may each have a circular shape around the active area **103**, in particular forming concentric circles with each other.

[0134] Compared to what has been shown and described for the device **100**, the pockets **332** may have, at the surface, variable spacing and widths.

[0135] The pockets **332** are distributed in the semiconductor body **110** in such a way that the edge termination region **330** has a non-uniform P-type pocket density (number of pockets per unit area or unit volume) inside the width $W_{sub.t}$.

[0136] In detail, the pockets **332** may have increasing density moving parallel to the X axis from the active area **103** towards the outer edge **102** of the die **101**. For example, the width of the pockets **332** and/or the distance along the X axis between two adjacent pockets **332** may have a decreasing trend moving away from the active area **103** parallel to the X axis and towards the outer edge **102**.

[0137] For example, as shown in FIG. **6**, two adjacent pockets **332** along the x axis may be arranged from each other at a distance W'_n in proximity to the active area **103**, a distance $W''_{sub.h}$ at a central portion of the edge termination region **330**, and a distance $W'''_{sub.h}$ in proximity to the edge **102**, where $W'''_{sub.h} < W''_{sub.h} < W'_{sub.h}$.

[0138] For example, the distance $W'.sub.h$ may be comprised between 3 μm and 8 μm . For example, the distance $W''.sub.h$ may be comprised between 2 μm and 7 μm . For example, the distance $W'''.sub.h$ may be comprised between 1 μm and 6 μm .

[0139] The edge termination region **330** allows the device **300** to have, in use, a high breakdown voltage.

[0140] In particular, the fact that the density of the pockets **332** is not uniform in the edge termination region **330**, and even more in particular the density is greater at greater distances from the active area **103**, may ensure high electrical performances.

[0141] FIG. 7 shows a further embodiment of the present electronic device, indicated here by **400**. The electronic device **400** has a general structure similar to that of the device **100**; accordingly, elements in common are indicated by the same reference numerals and are not further described in detail.

[0142] The device **400** is also a vertical-conduction MOSFET device of the superjunction type.

[0143] In detail, the device **400** comprises, in the active area **103**, the surface body regions **115**, the deep body regions **116**, the source regions **117** and the gate regions **118**.

[0144] The device **400** comprises, in the edge region **104**, an edge termination region **430** which extends inside the semiconductor body **110** around the active area **103**.

[0145] The edge termination region **430** comprises a plurality of pockets **432**, also here of an opposite conductivity type (having a P-type doping) with respect to that of the semiconductor body **110** (having an N-type doping), extending into the semiconductor body **110**.

[0146] The pockets **432** may have, around the active area **103**, in plan view on planes parallel to the XY plane, the same trend as discussed with reference to the pockets **132** of FIG. 3, for example so as to each form a circle around the active area **103**. The circles formed by the pockets **332B** of each row may also here be concentric with each other.

[0147] In detail, the plurality of pockets **432** includes surface pockets **432A** extending at the front surface **110A**, and deep pockets **432B** extending at a distance from the front surface **110A**.

[0148] The edge termination region **430** has a width, measured along a direction which extends between the active area **103** and the outer edge **102** (X axis in FIG. 7), variable as a function of the depth in the semiconductor body **110**. In particular, the width is decreasing moving from the front surface **110A** towards the rear surface **110B** along the Z axis.

[0149] In detail, at the front surface **110A**, the surface pockets **432A** define an upper width $W.sub.t,f$ of the edge termination region **430**, comprised for example between 50 μm and 200 μm .

[0150] The deep pockets **432B** define, in depth in the epitaxial region **112**, a rear width $W.sub.t,b$ of the edge termination region **430**, comprised for example between 30 μm and 120 μm .

[0151] The upper width W_{tf} is greater than the rear width $W.sub.t,b$; in particular the upper width W_{tf} defines a maximum width of the edge termination region **430** and the rear width $W.sub.t,b$ defines a minimum width of the edge termination region **430**.

[0152] Furthermore, the pockets **432** may be arranged in such a way that the edge termination region **430** has a non-uniform density of the pockets **432**, for example greater at greater depths within the epitaxial region **112** from the front surface **110A**. This may ensure a high breakdown voltage, in use, of the device **400**.

[0153] For example, moving from the front surface **110A** towards the rear surface **110B** parallel to the Z axis, the pockets **432** may have a width, measured along the X axis, that is decreasing.

[0154] Electrical simulations performed by the Applicant have verified that the decreasing trend of the width of the edge termination region **430** may be useful to ensure, in use, excellent electrical performance of the device **400**.

[0155] In practice, with reference to the devices **100**, **300**, **400**, the fact that the respective edge termination region **130**, **330**, **430** is formed by a plurality of pockets **132**, **332**, **432** having a conductivity type different from that of the semiconductor body **110** and arranged at a distance from each other along the Z axis or along both the Z axis and along the X axis, gives a high design

versatility of the edge termination region. The present electronic device may therefore be adapted to a high variety of applications.

[0156] It is also clear to the person skilled in the art that the manufacturing steps described with reference to FIGS. 5A to 5G may be adapted for the manufacturing of the devices **300**, **400**.

[0157] FIGS. **8** and **9** show further embodiments of the present semiconductor electronic device, indicated by **500** and **600**, respectively.

[0158] The devices **500** and **600** comprise, in the edge region **104** around the active area, an edge termination region having P-type pockets (indicated by **502** and **602**, respectively).

[0159] The pockets **502**, **602** may have a distribution in the semiconductor body **110** that is equal to or different from what has been discussed for the pockets **132** (FIG. **4**), **332** (FIG. **6**) and **432** (FIG. **7**).

[0160] In particular, the devices **500** and **600** differ from the device **100** of FIGS. **3** and **4** in the distribution of the pockets **502**, **602** around the active area **103**.

[0161] In detail, in the devices **500** and **600**, the pockets **502**, **602** extend discontinuously around the active area **103**, thus forming a plurality of portions separated from each other.

[0162] The discontinuous distribution of the pockets **502** or **602** around the active area **103** may simplify the design step, in particular of the lithography masks, of the devices **500** and **600**.

[0163] Finally, it is clear that modifications and variations may be made to the semiconductor electronic devices and the manufacturing processes thereof described and illustrated herein without thereby departing from the scope of the present invention, as defined in the attached claims.

[0164] For example, the surface pockets and/or the deep pockets of the edge termination regions may be (all or only some of them) floating or have dedicated regions for biasing to a specific electric potential. In particular, the surface pockets and the deep pockets may be (for example, all) floating; in this case, the actual potential of the pockets may be established in the design step through appropriate sizing as a function of the specific application. Furthermore, using floating pockets allows the manufacturing of the device to be simplified.

[0165] For example, the edge termination region may have a different pocket number and distribution than described.

[0166] For example, the deep body regions and/or the doped pockets of the edge termination region may be manufactured, alternatively or in addition to subsequent steps of epitaxial growth and implantation of doping species, by forming trenches in the wafer **200** and filling the trenches with doped semiconductor material.

[0167] For example, the gate regions **118** may be of the trench type (i.e., extending along the Z axis into the semiconductor body **110**).

[0168] For example, the active area **103** may also accommodate elementary cells of devices other than a MOSFET, according to the specific application of the semiconductor electronic device.

[0169] For example, the present invention also applies to a semiconductor electronic device other than a vertical-conduction MOSFET of the superjunction-type, such as for example lateral conduction MOSFETs, vertical-conduction MOSFETs and not of the superjunction-type, or devices other than a MOSFET such as trenchFETs, diodes, triristors, MESFETs, MISFETs, IGBTs, etc.

[0170] The conductivity types, P and N, may be reversed with respect to what has been discussed above. For example, the semiconductor body may be of a P-type and the pockets of the edge termination region may be of an N-type.

[0171] Finally, the different embodiments described and illustrated above may be combined to provide further solutions.

Claims

1. A semiconductor electronic device, comprising: a semiconductor body having a first conductivity type, a front surface and a rear surface extending at a distance from the front surface along a first

direction, the semiconductor body also having a lateral edge; an active area configured to include a conductive channel of the semiconductor electronic device; and an edge termination region extending around the active area between the active area and the lateral edge of the semiconductor body along a second direction transversal to the first direction; wherein the edge termination region comprises a plurality of doped portions having a second conductivity type different from the first conductivity type and arranged in the semiconductor body at a distance from each other either along the first direction or along both the first direction and the second direction.

2. The semiconductor electronic device according to claim 1, wherein the plurality of doped portions comprises surface doped portions extending into the semiconductor body starting from the front surface, wherein the surface doped portions are spaced at a distance from each other along the second direction.

3. The semiconductor electronic device according to claim 1, wherein the plurality of doped portions comprises deep doped portions extending into the semiconductor body at a distance from the front surface, and wherein the deep doped portions are spaced at a distance from each other along both the second direction and the first direction.

4. The semiconductor electronic device according to claim 1, wherein the plurality of doped portions comprises deep doped portions extending into the semiconductor body at a distance from the front surface, and wherein the deep doped portions are spaced at a distance from each other along the first direction.

5. The semiconductor electronic device according to claim 1, wherein the doped portions face each other and are aligned with each other along the first direction.

6. The semiconductor electronic device according to claim 1, wherein the doped portions face each other at least in part along the first direction.

7. The semiconductor electronic device according to claim 1, wherein the edge termination region has a uniform density of doped portions.

8. The semiconductor electronic device according to claim 1, wherein the edge termination region has a non-uniform density of doped portions.

9. The semiconductor electronic device according to claim 8, wherein the edge termination region has, in proximity to the active area, a first density of doped portions and, in proximity to the lateral edge of the semiconductor body, a second density of doped portions which is greater than the first density.

10. The semiconductor electronic device according claim 1, wherein the edge termination region has a density of doped portions increasing along the second direction, from the active area towards the lateral edge.

11. The semiconductor electronic device according to claim 1, wherein the edge termination region has a density of doped portions increasing along the first direction from the front surface towards the rear surface.

12. The semiconductor electronic device according to claim 1, wherein the edge termination region has a width along the second direction that is decreasing along the first direction from the front surface towards the rear surface.

13. The semiconductor electronic device according to claim 1, wherein the doped portions each have a width along the second direction comprised between 1 μm and 10 μm .

14. The semiconductor electronic device according to claim 1, wherein the device is a vertical-conduction MOSFET device.

15. The semiconductor electronic device according to claim 1, wherein the device is a superjunction MOSFET device having at least one deep body region extending into the active area at a distance from the front surface along the first direction.

16. A process for manufacturing a semiconductor electronic device, comprising: providing a wafer of semiconductor material having a first conductivity type, a front surface and a rear surface at a distance from the front surface along a first direction, the wafer further having a lateral edge;

forming an active area configured to include a conductive channel of the semiconductor electronic device; and forming an edge termination region extending around the active area, between the active area and the lateral edge of the wafer along a second direction transversal to the first direction; wherein forming an edge termination region comprises forming a plurality of doped portions, having a second conductivity type different from the first conductivity type, in the semiconductor body and at a distance from each other either along the first direction or along both the first direction and the second direction.

17. The process according to claim 16, wherein forming the edge termination region comprises performing at least once: growing at least one epitaxial layer on the front surface of the wafer; and forming the doped portions in the at least one epitaxial layer.

18. The process according to claim 16, further comprising forming, in the wafer, at least one deep body region having the second conductivity type, the deep body regions being formed, at least in part, simultaneously with the doped portions of the edge termination region.
